SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai 2026

ASSESSMENT TOOL -2

CO1_AT2

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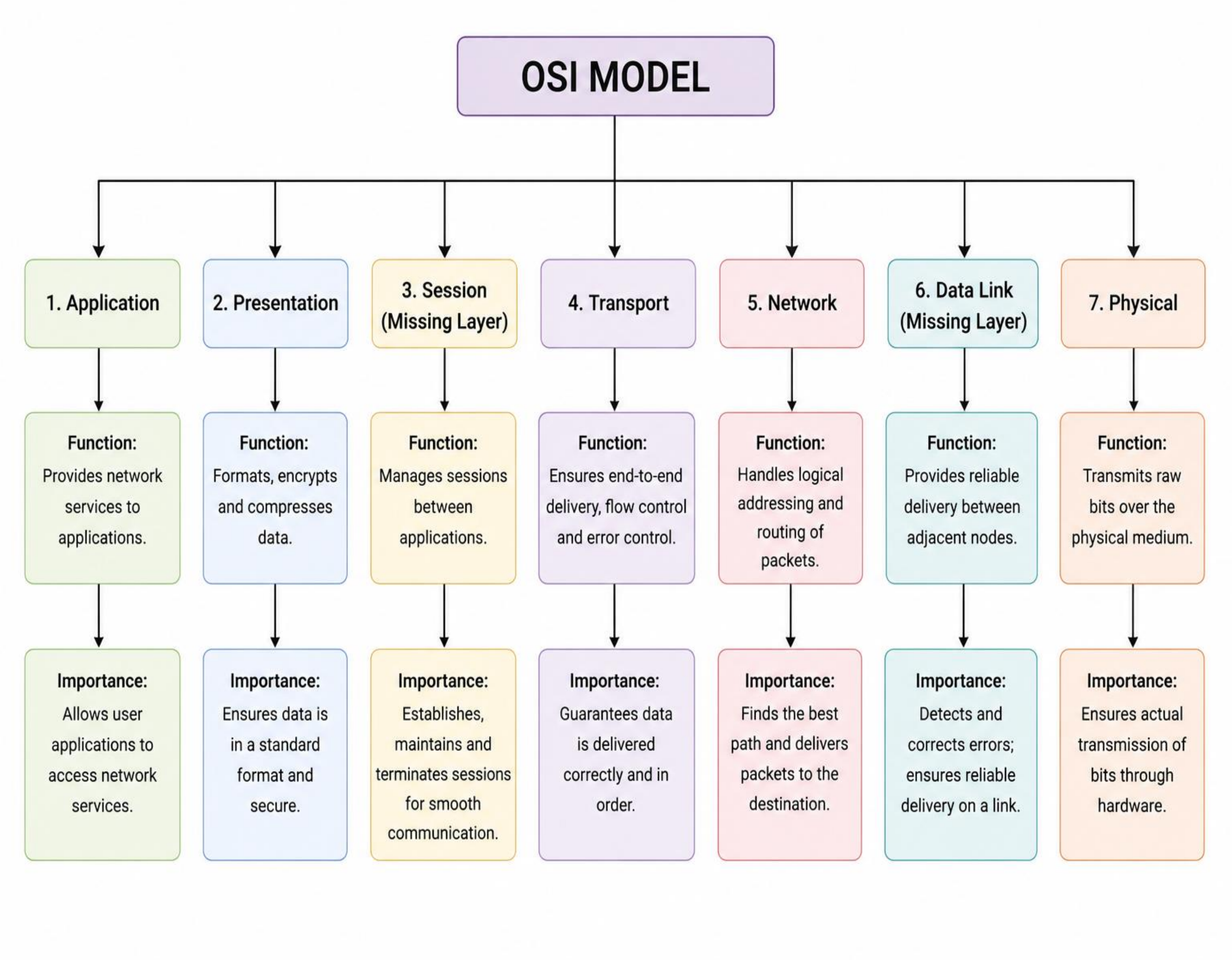
CONCEPT MAPPING

1.A partially completed concept map is given below:

Application → Presentation → ________ → Transport → Network → ________ → Physical. Analyze the missing layers and explain how their functions are essential for reliable communication.

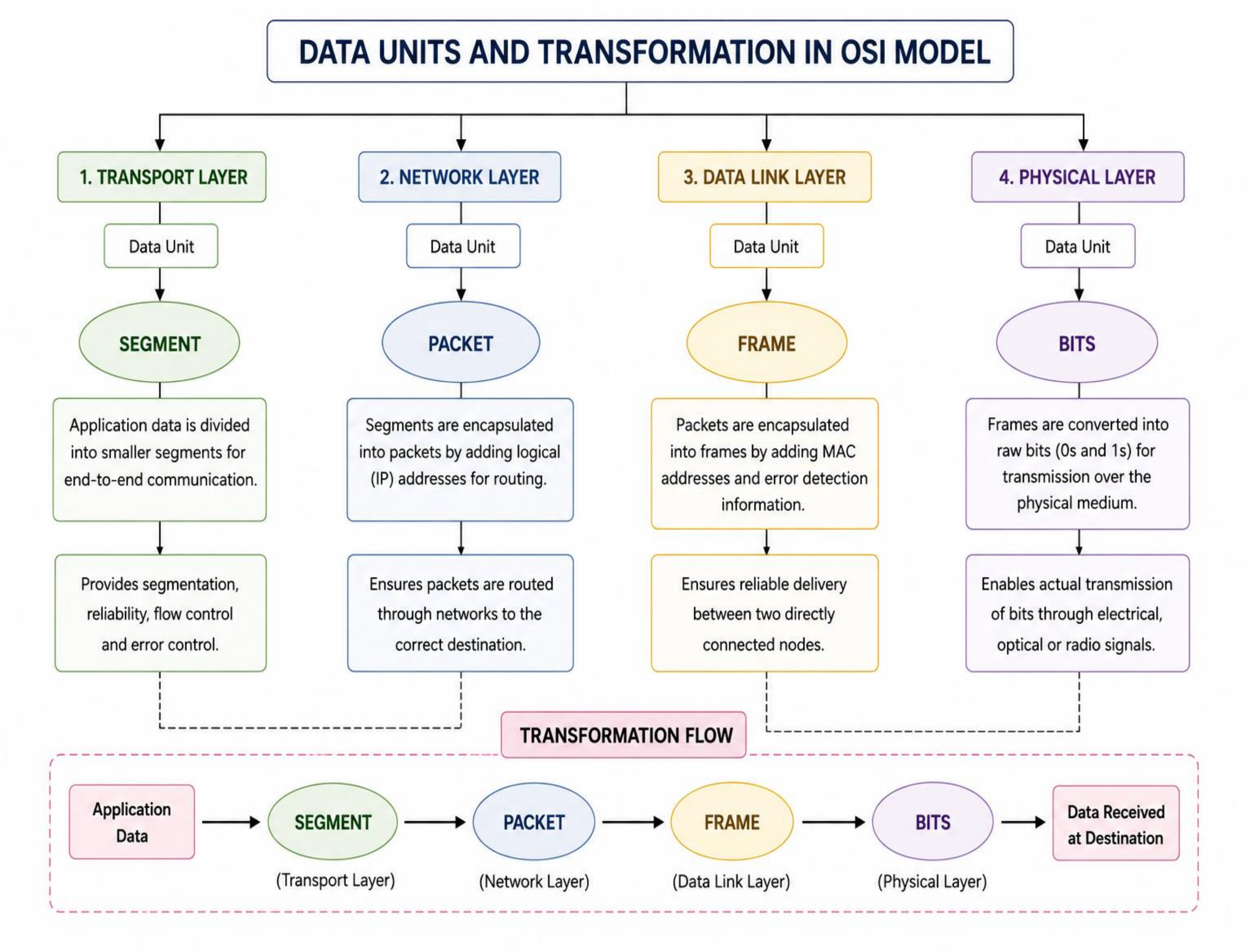
ANSWER :



- The missing layers are Session Layer and Data Link Layer.
- Session Layer establishes, manages, and terminates communication sessions.
- Data Link Layer provides node-to-node delivery and error detection.
- Both layers support reliable and organized communication between devices.
- Without these layers, data transfer becomes unreliable and communication may fail.

2.Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

ANSWER :



- Transport Layer uses Segments for end-to-end delivery.
- Network Layer converts segments into Packets for routing.
- Data Link Layer encapsulates packets into Frames for local delivery.
- Physical Layer transmits frames as Bits over transmission media.
- The transformation of data units ensures accurate, efficient, and reliable transmission.

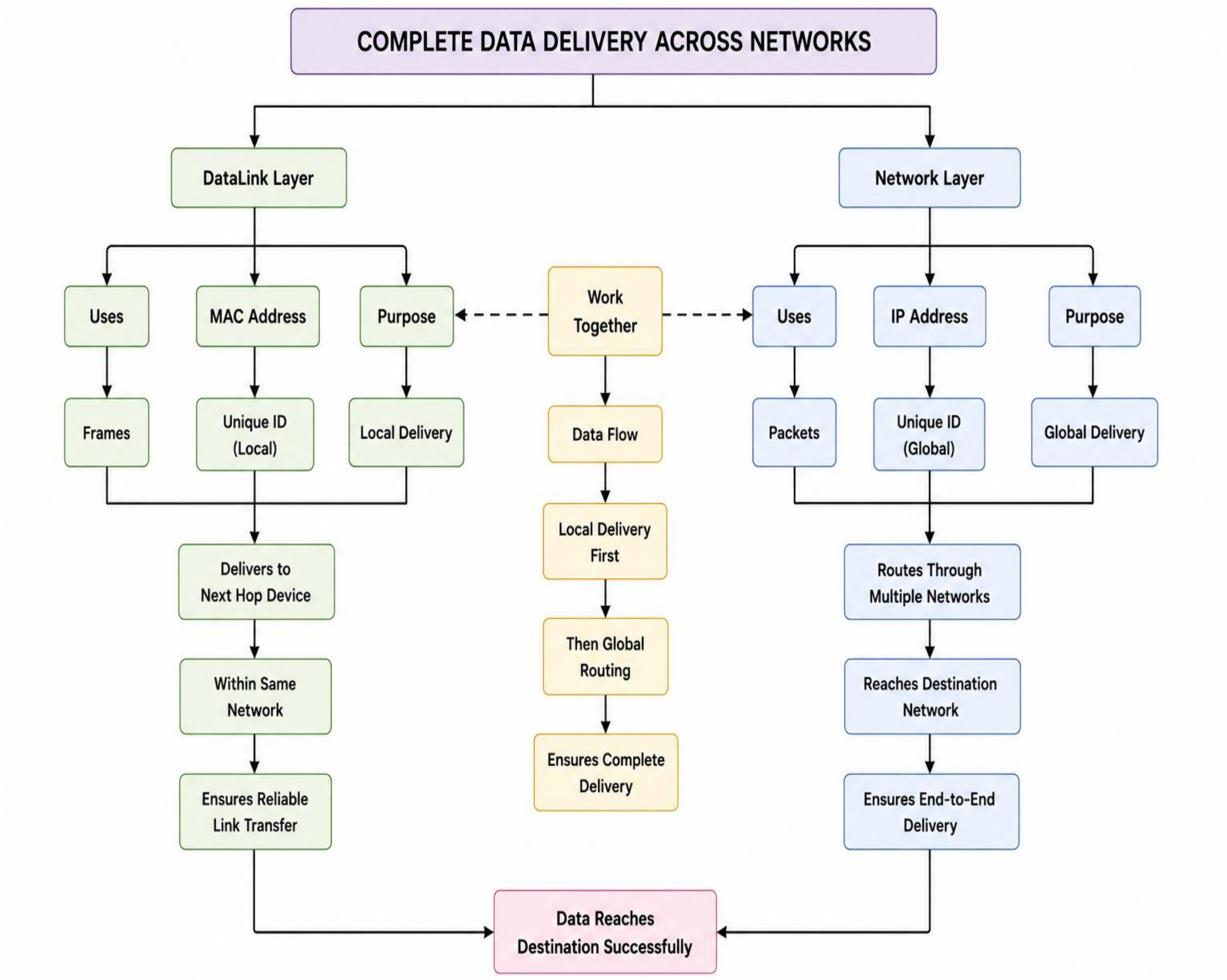
3.Given the concept map:

Data Link → MAC Address → Local Delivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

ANSWER :



- MAC Address enables local communication within the same network.
- IP Address enables communication across different networks.
- Data Link Layer uses MAC addresses for node-to-node delivery.
- Network Layer uses IP addresses for packet routing to destinations.
- Together, MAC and IP addresses ensure complete end-to-end data delivery.

4.A concept map shows:

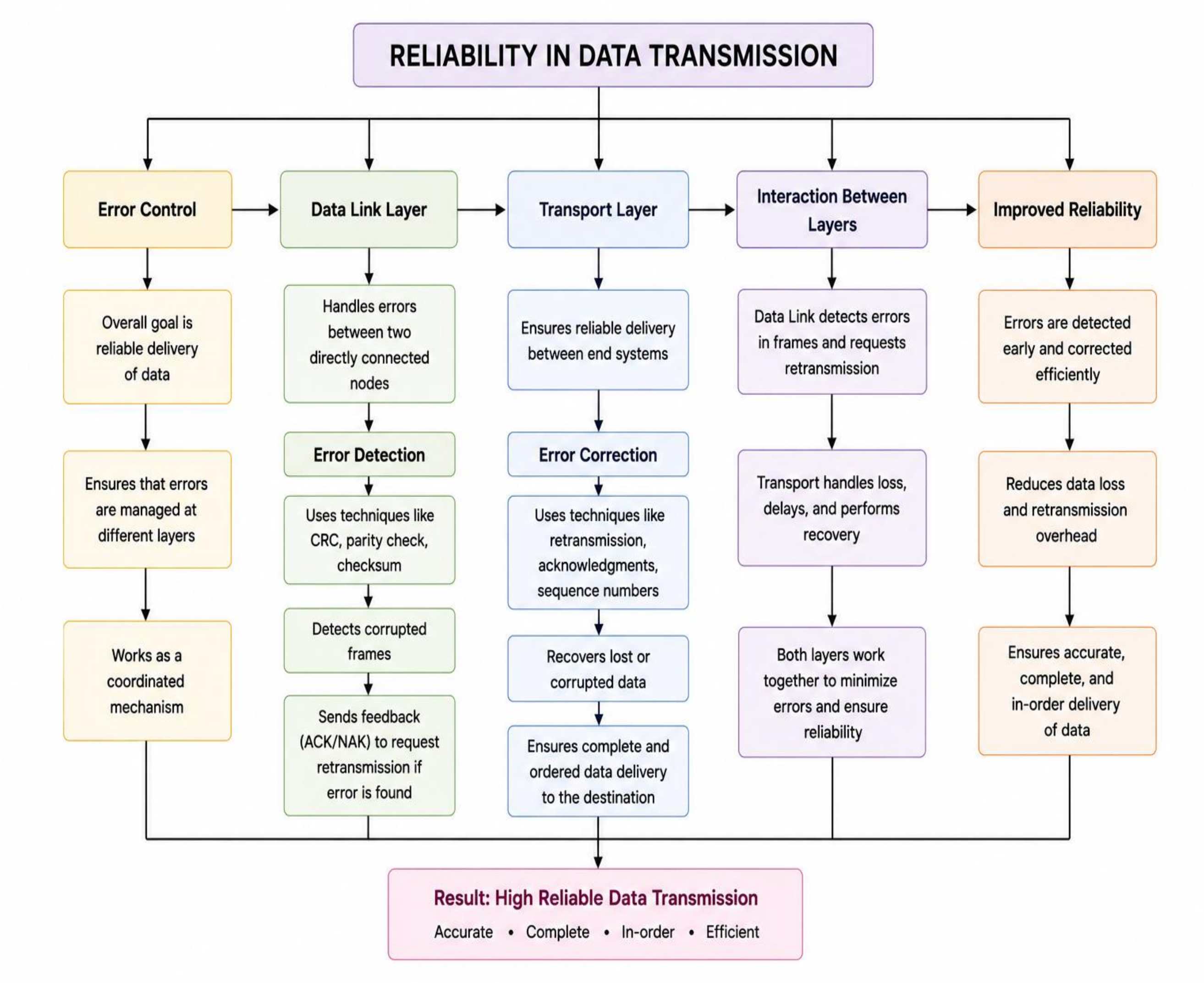
Error Control

→ Data Link Layer → Error Detection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

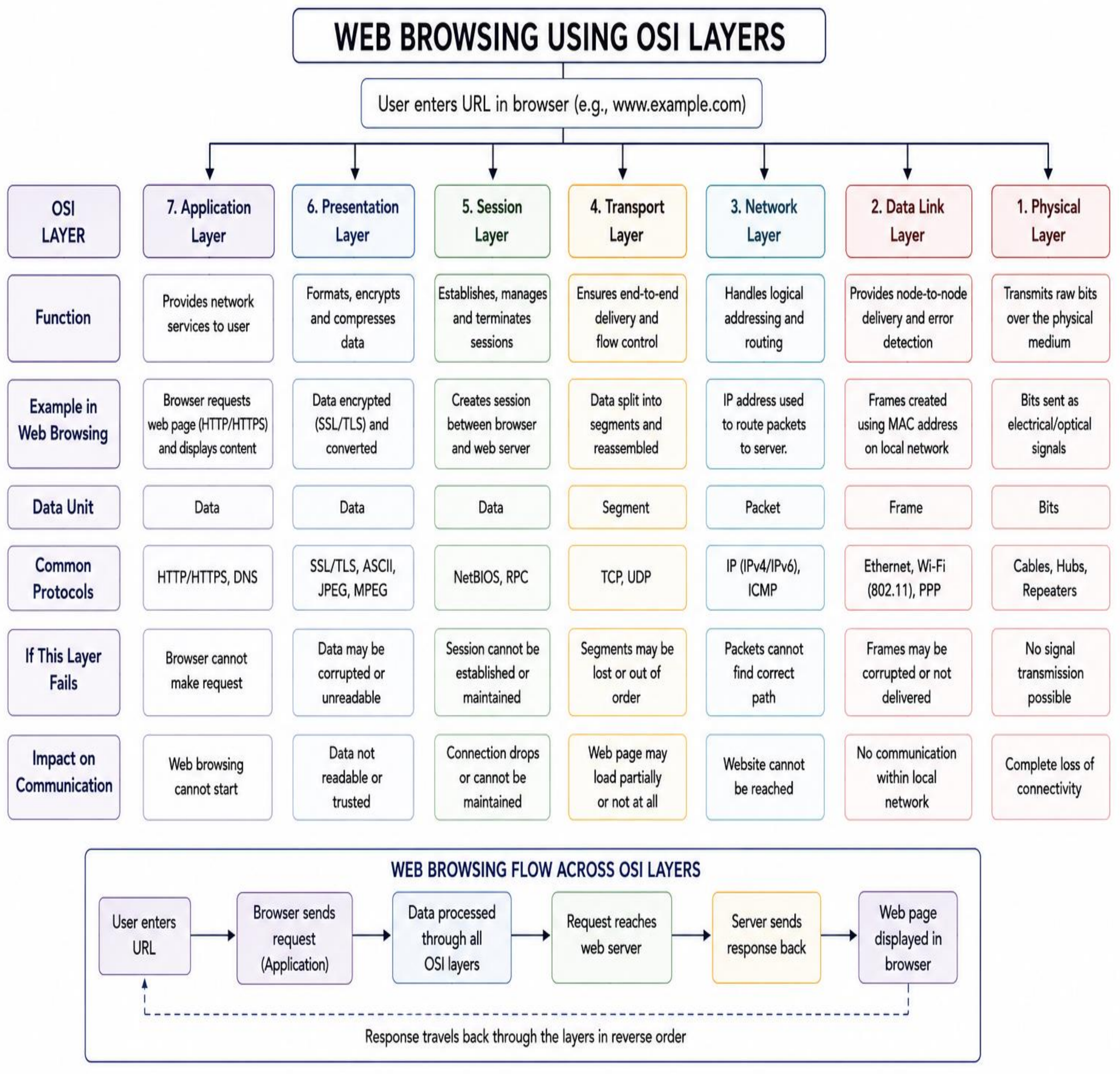
ANSWER :



- Data Link Layer performs error detection using techniques like CRC.
- Transport Layer performs error correction and retransmission.
- Data Link Layer identifies corrupted frames during transmission.

5.Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

ANSWER :



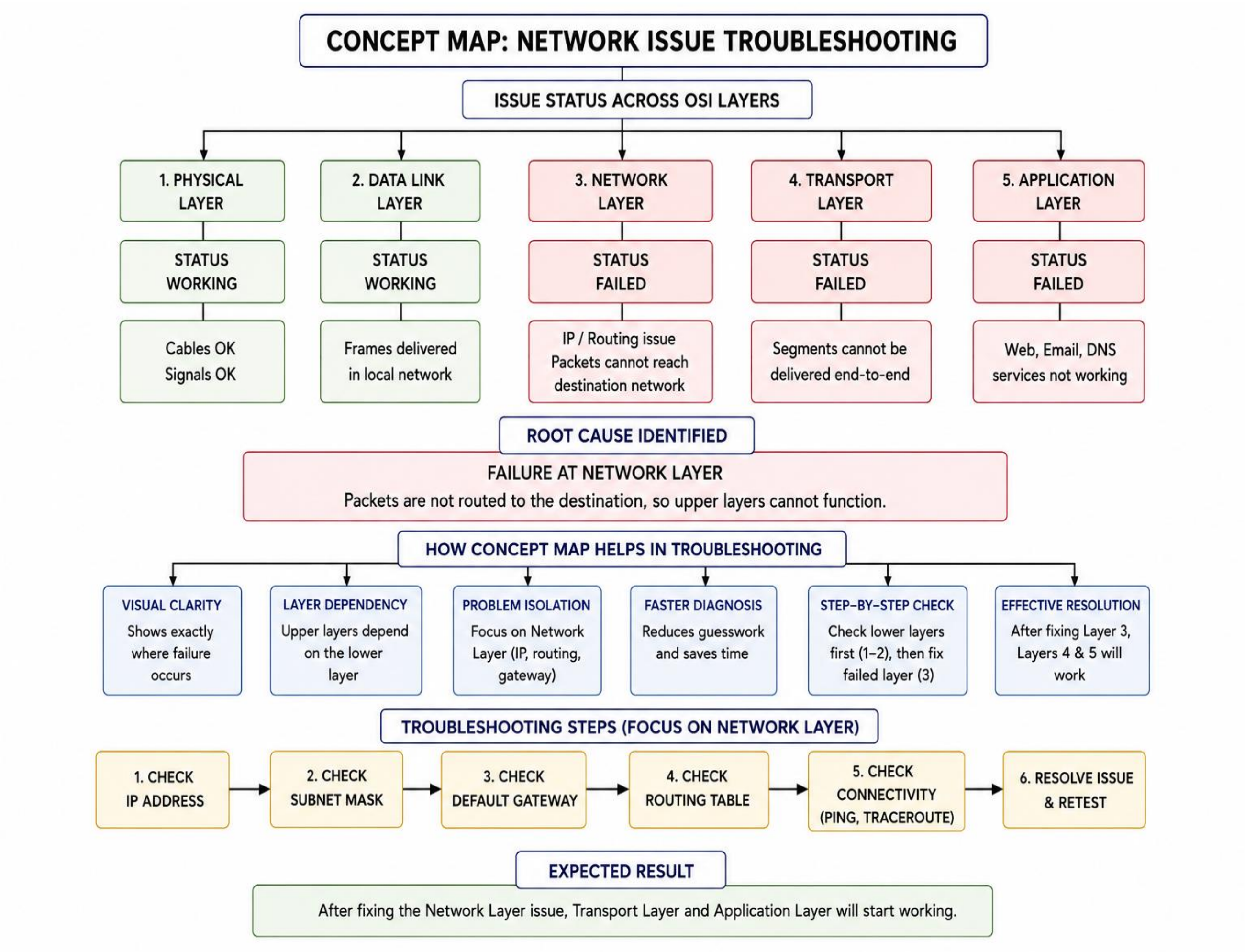
- Application Layer provides web services such as HTTP and HTTPS.
- Transport Layer ensures reliable delivery of web data using TCP.
- Network Layer routes packets between the browser and web server.
- Data Link and Physical Layers handle local delivery and signal transmission.
- Failure in any OSI layer interrupts the entire web browsing process.

6.A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

ANSWER :



- Physical and Data Link Layers are functioning correctly.
- The Network Layer has failed due to IP addressing or routing issues.
- Application services such as web browsing and email stop working.
- The concept map helps identify the failed layer quickly and simplifies troubleshooting.